

Reactions of Carboxylic Acids (Cambridge (CIE) A Level Chemistry): Revision Note

Reactions of Carboxylic Acids to Produce Acyl Chlorides

- **Acyl chlorides** are compounds with the functional group -COCl
- They look similar in structure to **carboxylic acids** but have a Cl atom instead of an -OH group attached to the carbonyl (C=O)
- Acyl chlorides are **more reactive** than their corresponding carboxylic acids and are therefore often used as **starting materials** in the production of organic compounds such as **esters**
- They can be prepared from the reaction of carboxylic acids with:
 - **Solid** phosphorus(V) chloride (PCl_5)
 - **Liquid** phosphorus(III) chloride (PCl_3) and heat
 - **Liquid** sulfur dichloride oxide (SOCl_2)
- For example, the acyl chloride **ethanoyl chloride** can be formed from **ethanoic acid** in the above reactions

Using ethanoic acid to form ethanoyl chloride



- **Methanoic acid** is a **strong reducing agent** and gets **further oxidised** to **carbon dioxide** (CO_2)
- The oxidation of methanoic acid can occur by:
 - Warming methanoic acid with **mild oxidising agents** such as **Fehling's** or **Tollens'** reagent
 - In a Fehling's solution, the Cu^{2+} ion is **reduced** to Cu^+ ion which **precipitates** as **red** Cu_2O

- With Tollens' reagent, the Ag^+ is **reduced** to Ag
- Using **stronger** oxidising agents such as **acidified KMnO_4** or **acidified $\text{K}_2\text{Cr}_2\text{O}_7$**
 - The **purple** KMnO_4 solution turns **colourless** as Mn^{7+} ions are **reduced** to Mn^{2+} ions
 - The **orange** $\text{K}_2\text{Cr}_2\text{O}_7$ solution turns **green** as the Cr^{6+} ions are reduced to Cr^{3+} ions

Ethanedioic acid

- Another carboxylic acid that can get further oxidised is **ethanedioic acid**
- A **strong oxidising agent** such as **warm acidified KMnO_4** is required for the oxidation of ethanedioic acid to **carbon dioxide**

Oxidation of ethanedioic acid

Ethanedioic acid is a dicarboxylic acid that can get further oxidised to carbon dioxide